
Research article

Safe discrimination of fractional, delayed, and latent memory beyond classical waveforms in a strong-Allee predator–prey model

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Abstract: Fitting a fractional order α does not identify a fractional mechanism: on a finite horizon, delayed feedback or a few hidden states reproduce similar trajectories. We ask when a controlled experiment can separate these mechanisms in a strong-Allee predator–prey system while keeping the prey above its Allee threshold. Among six classical excitation families the accuracy-leading designs all cross the threshold and the only safe one is the least informative, which suggests an intrinsic safety–informativeness frontier. We show that frontier belongs to the waveform parameterisation, not to the system. Reducing amplitude does not restore safety — the leading design still crosses in 71% of cells at half amplitude — whereas a constrained search over 163 840 candidate waveforms, at the same peak-amplitude budget and under a hard margin constraint, returns piecewise-constant designs with four-class model-selection accuracy 0.803 and no observed crossings, against 0.514 for the only safe classical family and 0.763 for the best classical design of any kind, which crosses in 36% of cells. Their safety is verified a posteriori by mesh refinement with a rigorous inter-node modulus, and for half of the headline trajectories by a validated enclosure. A Pareto cost nevertheless remains, so safety is not free. A second obstruction survives waveform search: computed at the prey-response level, the finite-state approximation error falls to $5 \cdot 10^{-5}$ at latent order 128 and the induced exact minimax testing floor reaches 0.49999. Identifying ecological memory is limited by the declared rival-complexity budget rather than by the safety margin.

Keywords: fractional-order dynamics; time-delay systems; model discrimination; experimental design; Allee effect; predator–prey dynamics

Mathematics Subject Classification: 34A08; 34K20; 62K05; 92D25

1. Introduction

Caputo derivatives of order $0 < \alpha < 1$ model hereditary effects in viscoelasticity, anomalous transport, and population dynamics [5, 6, 12, 19]; fractional predator–prey models in particular form a sizeable literature [1, 10, 20, 21]. An empirical difficulty sits underneath all of it. Estimating $\alpha < 1$ from data does not establish a fractional mechanism, because on a bounded window $[0, T]$ a delay term $\xi(t - \tau)$ or a low-dimensional latent state $q \in \mathbb{R}^m$ generates trajectories within the noise floor of the fractional ones (Corollary 3.3 makes this quantitative). Design methods for fractional-order *estimation* exist [8, 9, 15, 23]; safety-constrained information acquisition has been analyzed in general settings [18, 22]. The intersection — discrimination between memory *mechanisms*, under an ecological safety constraint — is the subject here:

Which perturbations and measurements can distinguish fractional memory from delayed or latent alternatives in a strong-Allee ecological system, and when does ecological safety make that discrimination fundamentally difficult?

Safety is not decorative in this question. Below the Allee threshold A the prey growth term $P(x) = rx(1 - x/K)(x/A - 1)$ turns negative and collapse is irreversible, so the perturbations most informative about memory ($\|u\|_\infty$ near the budget, broadband spectra) are the ones most likely to destroy the object of study. Section 8 measures this numerically: among the six classical excitation families, the best-discriminating input crosses the threshold in 100% of stable-regime cells. Section 8.4 then shows that this apparent conflict is a property of those families rather than of the ecology.

Provenance of the ecological backbone. The ecological backbone follows the companion certification study [14]: the equations, locked parameters, coexistence equilibrium and Jacobian are restated here for self-containment and are not claimed as contributions of the present work. That study certifies dynamical regimes of a specified Caputo model; it does not address whether such a model can be told apart from a delayed or latent mechanism reproducing the same observations to within σ , which is the question posed here. No result below recertifies the backbone. With the ecology frozen, differences between simulated observations are attributable to the memory law or to the experiment, never to an altered population model. Only the integer-order and Caputo mechanisms share $J(A)$ exactly; the delayed and latent rivals are calibrated alternative response laws matched at the same operating point, input channel and units.

Main contributions. The paper is organised around one falsification and one positive design result, with a separate obstruction that survives both.

1. **Safe waveforms outside the classical families outperform every classical baseline** (Section 8.4). Under the same peak-amplitude budget $\|u\|_\infty \leq 0.100$ and a hard Allee-margin constraint, a constrained search over 163 840 candidate waveforms returns piecewise-constant designs with four-class BIC macro-accuracy 0.803 and zero observed threshold crossings, against 0.514 for the only classical family that is safe at this budget and 0.763 for the best classical design of any kind, which crosses in 35.7% of cells. The gain is concentrated where the safe classical design fails: Caputo recall rises from 0.404 to 0.911, delay recall from 0.445 to 0.853.
2. **The six-family safety-informativeness frontier is not a property of the system** (Sections 8.3–8.4). Reducing amplitude does not restore safety — the leading design still crosses in 71.4% of cells at half amplitude, while its realised response gain rises to 4.86 times the linear certificate, the signature of basin escape rather than linear response. The severe frontier reported for classical excitation families is therefore a statement about that parameterisation. A Pareto cost remains: the best unsafe candidate separates the hypotheses substantially better than the best safe one, so safety is not free.
3. **A separate finite-horizon latent-complexity obstruction, quantified at response level to $m = 128$** (Sections 3.3–3.5). Computed directly on the prey response rather than inferred from the memory kernel, the L^1 surrogate error falls to $4.8 \cdot 10^{-5}$ at $m = 128$ and the induced exact minimax testing floor reaches 0.49999: at attainable latent orders the finite-state rival is indistinguishable under the declared protocol, for every input in the budget. Waveform search does not remove this. In the benchmark the leading safe design also recovers only 0.475 recall against the finite-latent rival, which is consistent with the obstruction but not evidence for it: the BIC penalty on the higher-dimensional model is an alternative explanation that this study does not separate.
4. **Trajectory safety status, stated exactly** (Sections 7 and 8). Safety of the reported designs is verified a posteriori by mesh refinement together with a rigorous inter-node modulus; for four of the eight headline trajectories a validated enclosure with self-consistent error control is obtained, and only those are called rigorous. Constants entering the nonlinear invariant-set route are interval-certified, and that route is shown to admit inputs 70 times smaller than the amplitude actually used.
5. **Reproducibility control.** The frozen four-class benchmark reproduces to four decimal places on an independent host under a re-implemented driver (2268 cells across three amplitudes, no solver divergences).

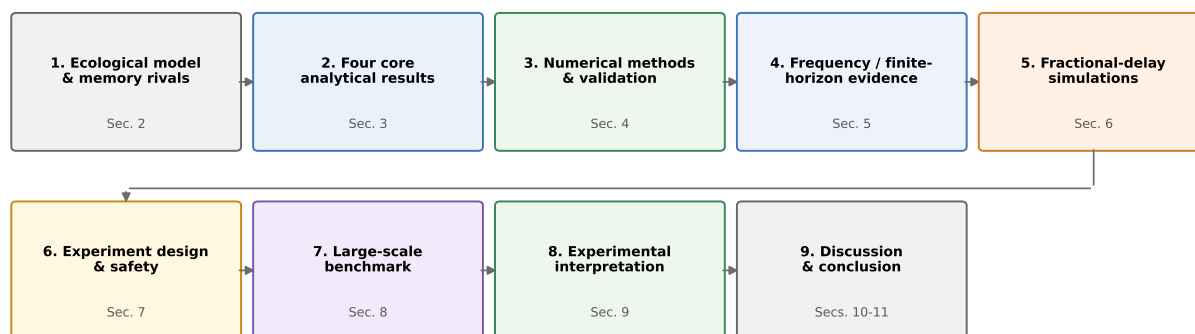
Contributions 1, 2 and 5 are empirical, established by the benchmark and the search; contribution 3 combines a proved bound (Theorem 3.4) with computed surrogate errors, interval-enclosed for $m \leq 32$ and high-accuracy numerical for $m = 64, 128$; contribution 4 is a verification result whose exact status is stated per trajectory in Section 8.

The structural separation of Section 3 and the positive exponential-sum approximation it is contrasted with are *not* contributions: the former is classical fractional-systems material, and the latter is the construction of McLean [17]

building on Beylkin and Monzón [3, 4]. We record only an endpoint-inclusive $L^1(0, T)$ variant with the exact horizon scaling the convolution argument requires (Lemma 3.2), and attribute the technique where it is used.

Complete proofs, secondary results, the Bayesian sequential extension, and solver derivations are in the Supplementary Material; the main text keeps the four statements needed to read the numerics.

Paper architecture: compact theory, expanded simulations, and a dedicated fractional-delay study



Theory is compacted in the main text; detailed proofs and the Bayesian extension are in the Supplement.

Figure 1. Structure of the paper. Sections 2–3: ecological model, rival memory laws, and the four analytical results. Sections 4–5: solver validation (Table 2) and frequency/finite-horizon evidence. Section 6: fractional-delay simulations. Sections 7–8: input design, safety, and the 1512-cell benchmark. Proofs and secondary theory: Supplementary Material.

2. Ecological Model and Competing Memory Mechanisms

2.1. Strong-Allee predator–prey backbone

All objects in this subsection are inherited from [14], where they were derived and certified with validated numerics: the model, the locked parameters, the equilibrium $(x^*, y^*(A))$, the Jacobian $J(A)$, and the Matignon boundary $\alpha^*(A)$. They are restated because the experiments of Sections 4–8 are defined in terms of them; none is claimed as new.

The nonlinear model is

$$f_1(x, y) = rx \left(1 - \frac{x}{K}\right) \left(\frac{x}{A} - 1\right) - \frac{axy}{1 + hx}, \quad (2.1)$$

$$f_2(x, y) = e \frac{axy}{1 + hx} - my, \quad (2.2)$$

with locked parameters

$$r = \frac{3}{2}, \quad K = 1, \quad a = 1, \quad h = \frac{1}{2}, \quad e = \frac{4}{5}, \quad m = \frac{2}{5}.$$

The coexistence equilibrium and Jacobian are

$$x^* = \frac{2}{3}, \quad y^*(A) = \frac{2(2 - 3A)}{9A},$$

$$J(A) = \begin{pmatrix} \frac{7A - 2}{8A} & -\frac{1}{2} \\ \frac{2 - 3A}{10A} & 0 \end{pmatrix}.$$

For the commensurate Caputo model, local stability follows Matignon’s sector condition $|\arg \mu| > \alpha\pi/2$ on the eigenvalues μ of $J(A)$ [16]. In the complex-pair regime $2/7 < A < 10/19$ the critical order is

$$\alpha^*(A) = \frac{2}{\pi} \arctan \left(\frac{\sqrt{4D(A) - T(A)^2}}{T(A)} \right),$$

where $T(A) = \text{tr } J$ and $D(A) = \det J$. At $A = 0.30$, $\alpha^* \approx 0.969012276$: the Caputo system with $\alpha = 0.9 < \alpha^*$ is stable while its integer-order limit ($\alpha = 1 > \alpha^*$) is unstable [14]. Most discrimination experiments below use $A \leq 2/7$, where every candidate mechanism is stable at the operating point and stability cannot serve as a discrimination shortcut.

Inherited backbone versus new experimental layer. Notation is kept identical to [14] for an identification reason: if the ecological equations changed between papers, a difference in simulated observations could reflect the altered biology instead of the memory law. The ecology is therefore frozen and only the response law varies — ${}^C D_t^\alpha$, the delay $\xi(t - \tau)$, or a latent state $q \in \mathbb{R}^m$ — with all candidates facing the same budget $\|u\|_\infty \leq 0.10$, horizon $T = 12$, and threshold A . Table 1 lists the frozen quantities.

Table 1. Locked ecological backbone used by both manuscripts. These values are inherited inputs, not new contributions of the present paper.

Quantity	Value	Role in present experiments
r	1.5	intrinsic prey growth
K	1	prey carrying capacity
a	1	attack-rate scale
h	0.5	Holling-II handling parameter
e	0.8	conversion efficiency
m	0.4	predator mortality
x^*	$2/3$	common coexistence prey level
$y^*(A)$	$2(2 - 3A)/(9A)$	coexistence predator level
$J(A)$	Eq. above	ODE/Caputo local ecological generator

The threshold A has two roles. First, $x(t) < A$ defines the failure event: $P(x) < 0$ there, and the trajectory can leave the coexistence basin permanently (Figure 3, left). Second, $x^* - A$ is the natural safety margin; at $A = 0.25$ it equals $2/3 - 0.25 \approx 0.417$. An input can thus score highly on the information criterion of Section 7 and still be ecologically inadmissible — Tables 10 and 7 quantify exactly this.

System, observation channels, and rival memory laws

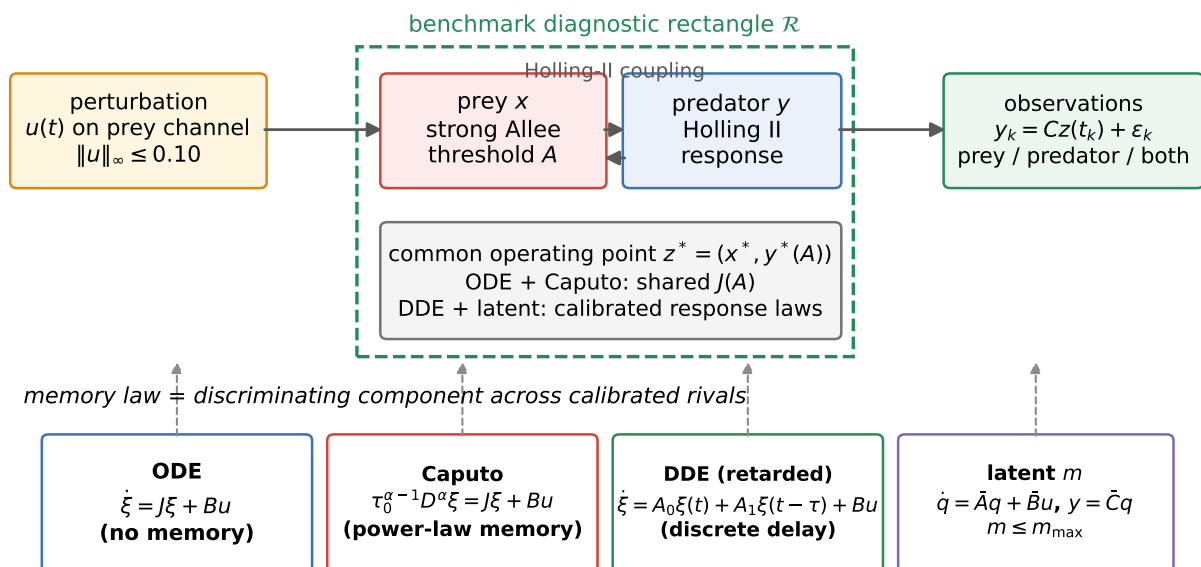


Figure 2. Setting: backbone from [14]; perturbation $u(t)$ on the prey channel with $\|u\|_\infty \leq 0.10$; observations $y_k = Cz(t_k) + \varepsilon_k$ on prey, predator, or both. ODE and Caputo share $J(A)$ exactly; DDE and latent are calibrated rival response laws. Dashed rectangle: numerical benchmark diagnostic region only (not an invariant-set certificate).

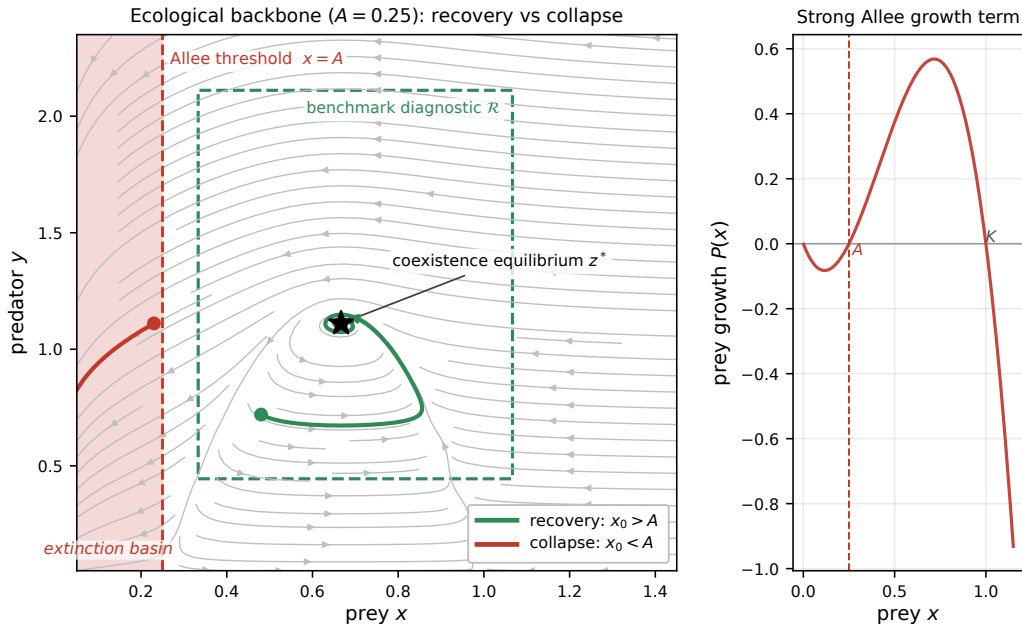


Figure 3. Safety geometry at $A = 0.25$ (backbone from [14]). Left: recovery for $x_0 > A$ versus collapse for $x_0 < A$. Right: $P(x) = rx(1 - x/K)(x/A - 1)$, with sign change at $x = A$.

2.2. Four rival mechanisms

With the reference time τ_0 enforcing dimensional consistency [6, 12, 19], the four candidate mechanisms around the operating point are

$$\begin{aligned} \text{ODE: } \dot{\xi} &= J\xi + Bu, \\ \text{Caputo: } \tau_0^{\alpha-1} D_t^\alpha \xi &= J\xi + Bu, \\ \text{DDE: } \dot{\xi} &= A_0 \xi(t) + A_1 \xi(t - \tau) + Bu, \\ \text{latent: } \dot{q} &= \bar{A}q + \bar{B}u, \quad y = \bar{C}q. \end{aligned}$$

The Caputo mechanism is the only one carrying a fractional order. The latent class, which would otherwise be flexible enough to imitate anything, is held to an explicit state-dimension budget m — a restriction whose necessity is precisely the content of Result II below.

Combined fractional-delay stress-test model. A system may carry hereditary memory and a trophic lag simultaneously, so we also simulate the combined model

$$\tau_0^{\alpha-1} D_t^\alpha x(t) = P(x(t); A) - \frac{ax(t)y(t)}{1 + hx(t)} + u(t), \quad (2.3)$$

$$\tau_0^{\alpha-1} D_t^\alpha y(t) = e \frac{ax(t - \tau)y(t)}{1 + hx(t - \tau)} - my(t), \quad (2.4)$$

with constant equilibrium history on $[-\tau, 0]$. Its boundaries are exact: at $\tau = 0$, (2.4) is the pure Caputo model; at $\alpha = 1$ it is the retarded DDE benchmark. Both limits are verified numerically to 6.75×10^{-6} and better in Table 2. The model is a numerical stress test only; no theorem depends on it.

3. Analytical Backbone: What Is Proved, What Is Borrowed, What Is Computed

This section fixes the mathematics the experiments rest on. Each item is labelled by status: *foundation* (standard, restated for self-containment, no priority claimed), *borrowed tool* (prior art, attributed), *proved here*, or *computed* — and, for computed quantities, whether the number is an interval enclosure or a high-accuracy floating-point evaluation. Three error objects recur and are never interchanged: the memory-kernel error, the induced response-operator error, and the fixed-input discrepancy (Section 3.3).

3.1. Result I: exact structural separation (foundation)

The collocated Caputo transfer of the linearisation at $(x^*, y^*(A))$ carries an $O(\omega^{-\alpha})$ high-frequency signature. No finite rational latent transfer (integer-order rolloff) and no strictly proper retarded-delay model ($O(\omega^{-1})$) reproduces it exactly on an open frequency set. The mechanism — a branch point of s^α at the origin against the meromorphy of the rival classes — is classical fractional-systems material [6, 13, 19]; no novelty is claimed for it. It is stated because it delimits what the later results can and cannot mean.

Theorem 3.1 (Structural separation). *Let $CB \neq 0$ and $\alpha \in (0, 1)$ with $\alpha \notin \mathbb{Q}$ in the sense that no effectively present exponent of s^α is an integer multiple of $1/\alpha$. Then G_α is not meromorphic in any neighbourhood of $s = 0$, hence differs from every finite rational transfer and from every strictly proper retarded-delay transfer.*

The statement concerns exact equality of transfers on open frequency sets. It promises nothing about experiments with finite bandwidth, finite horizon $T < \infty$ and noise $\sigma > 0$. That regime is governed by Lemma 3.2 and its corollary.

3.2. Result II: positive exponential-sum approximation (borrowed tool)

The Caputo kernel $k_\alpha(t) = t^{\alpha-1}/\Gamma(\alpha)$ has the Stieltjes representation $k_\alpha(t) = c_\alpha \int_0^\infty \lambda^{-\alpha} e^{-\lambda t} d\lambda$ with $c_\alpha = \sin(\pi\alpha)/\pi$. Discretising it by the trapezoidal rule on a logarithmic grid $\lambda_k = e^{kh}$ and truncating produces an exponential sum with *positive* weights and rates.

Attribution. This construction is prior art. Positive exponential-sum approximation of $t^{-\beta}$ by quadrature of an integral representation, including the positivity of the weights and the root-exponential order in the number of terms, is due to McLean [17], building on the exponential-sum approximation theory of Beylkin and Monzón [3, 4]; the same construction underlies the fast Caputo evaluation of Jiang et al. [11]. The quadrature estimate we invoke is the analytic-strip trapezoidal bound [25], and root-exponential order is the expected optimal order for an algebraic branch point [24]. We claim no new approximation method, no new rate law, and no priority for positive weights.

What we add is narrow and technical: the published estimates are stated on a compact interval $[\delta, T]$ with $\delta > 0$, excluding the singularity at the origin, whereas the discrimination argument of Section 3.4 needs the convolution norm $L^1(0, T)$ including $t = 0$. The strip bound is t -integrable there because $\alpha > 0$, so no excision is required.

Lemma 3.2 (Endpoint-inclusive L^1 estimate and exact horizon scaling). *Let $E_m^+(\alpha, T)$ denote the least $L^1(0, T)$ error of an m -term exponential sum with positive weights and rates. Then*

$$E_m^+(\alpha, T) = T^\alpha E_m^+(\alpha, 1), \quad (3.1)$$

and for every $c < \pi\sqrt{\alpha(1-\alpha)}$ there exists $A(\alpha, c) < \infty$ with

$$E_m^+(\alpha, T) \leq A(\alpha, c) T^\alpha e^{-c\sqrt{m}} \quad (m \geq 1). \quad (3.2)$$

Consequently $m(\varepsilon) = O(\log^2(1/\varepsilon))$ terms suffice for accuracy ε .

Proof. Equation (3.1) is exact: substituting $t = Ts$ and using $k_\alpha(Ts) = T^{\alpha-1}k_\alpha(s)$ turns the $L^1(0, T)$ error into T^α times the $L^1(0, 1)$ error, and $(c_j, \lambda_j) \mapsto (T^{1-\alpha}c_j, T\lambda_j)$ is a bijection of the admissible positive mixtures. For (3.2) put $\lambda = e^x$, so $k_\alpha(t) = c_\alpha \int_{\mathbb{R}} g(x, t) dx$ with $g(x, t) = e^{(1-\alpha)x} e^{-te^x}$. For $|\operatorname{Im} x| \leq d < \pi/2$ one has $|g| \leq e^{(1-\alpha)x} e^{-te^x \cos d}$, whence the strip norm obeys $N(g(\cdot, t), d) \leq 2(\cos d)^{\alpha-1} \Gamma(1-\alpha) t^{\alpha-1}$. Integrating the trapezoidal estimate of [25] over $t \in (0, 1)$ and using $\int_0^1 t^{\alpha-1} dt = 1/\alpha$ together with $c_\alpha \Gamma(1-\alpha) = 1/\Gamma(\alpha)$ bounds the discretisation error by $2\Gamma(\alpha+1)^{-1} (\cos d)^{\alpha-1} e^{-2\pi d/h} (1 - e^{-2\pi d/h})^{-1}$; the truncation tails contribute $O(e^{-\alpha M_+ h})$ and $O(e^{-(1-\alpha)M_- h})$. Equalising the three exponents gives $m \approx 2\pi d h^{-2} [\alpha(1-\alpha)]^{-1}$ and exponent $\sqrt{2\pi d \alpha(1-\alpha)m}$, which tends to $\pi\sqrt{\alpha(1-\alpha)m}$ as $d \uparrow \pi/2$ while $(\cos d)^{\alpha-1} \rightarrow \infty$. Hence any c strictly below the limit is attained with a finite constant. $\square$

The quantifier order in (3.2) is not cosmetic: the constant diverges as c approaches $\pi\sqrt{\alpha(1-\alpha)}$, so the boundary rate is *not* established, and whether it is attained or optimal is open. Equation (3.1) is verified to machine precision over a 285-cell sweep ($\alpha \in \{0.30, \dots, 0.95\}$, $T \in \{1, 10, 100\}$, $m \leq 128$): the measured ratio $E(T)/E(1)$ equals T^α to all printed digits, and the relative error is T -invariant. Over the same sweep, (3.2) holds at $c = \pi\sqrt{\alpha(1-\alpha)}$ with constants between 0.29 and 11.1, and is attained to within a factor 1.00 at $\alpha = 0.70$, $m = 128$.

Corollary 3.3 (Finite-horizon closure of the latent hierarchy). *For every $T < \infty$ and $\varepsilon > 0$ some finite positive exponential mixture lies within ε of k_α in $L^1(0, T)$. Hence, absent a declared bound on the latent dimension, no fixed finite-horizon experiment can guarantee a uniform positive separation from the whole latent hierarchy: discrimination is meaningful only relative to a budget m .*

Results I and II together fix the geometry: exact separation (Theorem 3.1) coexists with ε -approximation at finite T (Corollary 3.3). In the benchmark of Section 8 the latent class is capped at $m = 3$; in the response-level computation of Section 3.5 the budget runs to $m = 128$.

3.3. Semantics: three error objects that must not be conflated

Write $G_\alpha(s) = C(s^\alpha I - J)^{-1}B$ for the exact prey-to-prey transfer and G_m for a strictly proper finite-state surrogate. Three distinct quantities appear below.

(O1) *Memory-kernel error* $E_m = \|k_\alpha - k_m\|_{L^1(0,T)}$: a statement about the fractional operator alone, independent of J, B, C . This is the object of Lemma 3.2.

(O2) *Induced response-operator error* $\hat{E}_m^{\text{state}} = \inf_{g_m} \|g_\alpha - g_m\|_{L^1(0,T)}$, the L^1 distance between impulse responses, i.e. the $L^\infty \rightarrow L^\infty$ operator norm of the difference of input–output maps. It is a *minimax* quantity: an infimum over rivals of a supremum over inputs.

(O3) *Fixed-input discrepancy* $\|(g_\alpha - g_m) * u\|/\|u\|$ for one specific u .

By construction (O3) $\leq$ (O2) for every admissible u . There is no inequality in either direction between (O1) and (O2), and no constant relating them: computed from the same surrogate family, the plant *attenuates* the kernel error monotonically, by a factor 1.009 at $m \approx 5$ falling to 0.675 at $m = 128$. Kernel-level numbers are therefore never propagated into response-level claims anywhere in this paper.

Two consequences are used later. First, because (O2) is a supremum over inputs, the testing bound of Section 3.4 is valid uniformly over the admissible input class, and conservative. Second, and for the same reason, $\hat{E}_m^{\text{state}}$ is an *obstruction certificate and not a design objective*: the frequency band that maximises $|G_\alpha - G_m|$ (here $\omega \approx 0.4$) selects the multiscale waveform, which excites 87% of the worst case at $m = 32$ and is nonetheless the *least* informative design in the four-class benchmark (Table 11). Maximising one pairwise discrepancy does not maximise multi-class accuracy.

3.4. Result III: testing obstruction at the response-operator level (proved here)

Fix the observation model: the prey channel is sampled at n times in $(0, T]$ and corrupted by additive Gaussian noise with common covariance $\sigma^2 I$ under both hypotheses. The two hypotheses are “the response is g_α ” and “the response is g_m ”, with means μ_α, μ_m obtained by convolving the respective response kernels with the same admissible input u . Because the covariance is common and known, the minimax error of the two-point test is available in closed form and no inequality is needed.

Theorem 3.4 (Exact two-point obstruction under a response-operator budget). *Let $\|u\|_2 \leq B$ and let $d = \|\mu_\alpha - \mu_m\|_2/\sigma$. Then the minimax error of any test between the two hypotheses satisfies*

$$P_e^* = \Phi\left(-\frac{d}{2}\right), \quad d \leq \frac{\hat{E}_m^{\text{state}} B}{\sigma}, \quad (3.3)$$

by Young’s inequality $\|(g_\alpha - g_m) * u\|_2 \leq \|g_\alpha - g_m\|_{L^1} \|u\|_2$. Hence

$$P_e^* \geq \Phi\left(-\frac{\hat{E}_m^{\text{state}} B}{2\sigma}\right) \longrightarrow \frac{1}{2} \quad \text{as } \hat{E}_m^{\text{state}} \rightarrow 0 \text{ with } B, \sigma \text{ fixed.} \quad (3.4)$$

Three points of hygiene. The bound is routed through the L^1 norm of the response kernel, never through $\|k_\alpha - k_m\|_{L^2}$, which is infinite for $\alpha \leq 1/2$ and would make any such statement vacuous on part of the declared range. The exact bound (3.3) is the primary statement; the Pinsker bound $\frac{1}{2}(1 - \sqrt{\text{KL}/2})$ is reported alongside in Table 4 only for comparability with the literature, and is uniformly looser — by 0.034 at $m = 4$.

3.5. Result IV: finite-state prey-response approximation (computed)

Results I–III concern operators. Result IV addresses the measured object, the prey-to-prey response g_α . The factorisation $(s^\alpha I - J)^{-1} = (sI - J)^{-1} s^{-\alpha}$ is false and is not used. Instead g_α is split by contour deformation into its principal-sheet pole pair — realised *exactly* by a stable two-state integer-order block — plus a branch-cut remainder, smooth on $(0, \infty)$, which is the only part requiring approximation.

Theorem 3.5 (Certified finite-state approximation of the prey response). *For the displayed Matignon-stable cells and $m \in \{4, 8, 16, 32\}$ there exists an explicit stable finite-state surrogate g_m with $\|g_\alpha - g_m\|_{L^1(0,T)} \leq \widehat{E}_m^{\text{state}}$, the bound being an outward-rounded interval enclosure over the admissible pole region and L^1 mass cap.*

The values of $\widehat{E}_m^{\text{state}}$ for $m \leq 32$ (Table 4) are **interval enclosures**. Table 5 continues the computation to $m = 64$ and $m = 128$ using one explicit positive-SOE surrogate rather than the optimised class; those numbers are **high-accuracy floating-point evaluations with a two-mesh quadrature-error estimate**, not enclosures, and because that surrogate is one admissible element rather than the infimum they are upper bounds on $\widehat{E}_m^{\text{state}}$ — so the floors they induce through (3.4) are valid and conservative. The distinction is maintained wherever these numbers are used.

Combining Theorem 3.4 with the computed values gives the operative conclusion: the testing floor rises from 0.374 at $m = 4$ to 0.4996 at $m = 64$ and 0.49999 at $m = 128$. At attainable latent orders the finite-state rival is indistinguishable from the fractional model under the declared protocol, for *every* input in the budget. This is the obstruction that the waveform search of Section 8.4 does not remove, and it is separate from the safety question entirely.

4. Numerical Methods and Validation

The computational layer supports the analytical claims of Section 3; it does not replace them. All scripts and the machine-readable outputs (JSON/NPZ) behind every figure are released with the paper, so each displayed number can be regenerated from source.

4.1. Solvers

ODE and latent-state systems are integrated by classical fourth-order Runge–Kutta with step $h = T/N$; retarded DDEs by an interpolated-history RK4 variant. Caputo systems use the fractional Adams–Bashforth–Moulton PECE scheme [19]; its scalar implementation is validated against the exact Mittag–Leffler solution $E_\alpha(\lambda t^\alpha)$, with maximum errors 8.70×10^{-4} , 4.64×10^{-4} , 2.52×10^{-4} at $N = 250, 500, 1000$ (Table 2).

The combined model (2.4) reuses the PECE history convolution and evaluates the delayed prey value $x(t - \tau)$ by linear interpolation of the already computed trajectory. Reported grids with $\tau > 0$ satisfy $\tau \geq h = T/N$, so the corrector never requires unknown history. The model’s two boundaries serve as validation gates:

- at $\tau = 0$ the code delegates to the validated Caputo solver (no duplicated computation);
- at $\alpha = 1$ the output is compared against the independent DDE RK4 solver.

At $A = 0.25$, $\tau = 0.35$, the $\alpha = 1$ discrepancy against the DDE solver is 6.75×10^{-6} ($N = 200$), 1.68×10^{-6} ($N = 400$), 4.20×10^{-7} ($N = 800$); the error falls by a factor of ≈ 4 per grid doubling. In the interior, at $(\alpha, \tau) = (0.85, 0.35)$, self-convergence against the $N = 800$ reference improves from 2.18×10^{-3} ($N = 200$) to 8.05×10^{-4} ($N = 400$).

4.2. Inputs, observations, and metrics

Six bounded input families are compared under the common peak budget $\|u\|_\infty \leq 0.10$: pulse, multiscale pulse train, sinusoid, multisine, chirp, and pseudorandom binary sequence (PRBS); see Figure 11. Observation channels: prey, predator, or both. Numerical safety is reported separately from solver survival via the minimum Allee margin

$$\min_t (x(t) - A)$$

and the fraction of runs crossing $x = A$.

Four additional diagnostics apply to the fractional-delay runs: the L^2 trajectory distance to the pure Caputo limit ($\tau = 0$), the distance to the pure DDE limit ($\alpha = 1$), the predator peak time and amplitude, and the bridge score $\min(d_{\text{Caputo}}, d_{\text{DDE}})$, which is small whenever *either* pure mechanism explains the data and large only when neither does.

4.3. Validation matrix and numerical resolution

Each numerical claim is tied to an explicit convergence gate rather than to a single solver run; Table 2 collects all eight. Three reference types appear: the exact Mittag–Leffler solution (Caputo gates), the independent DDE RK4 solver ($\alpha = 1$ boundary), and an $N = 800$ self-convergence reference (interior cell). All eight error sequences decrease monotonically in N .

Table 2. Numerical validation gates. The Caputo column uses the exact Mittag–Leffler reference; the fractional-delay boundary uses an independent DDE RK4 solution.

Gate	grid size	Reference	max. abs. discrepancy
Caputo PECE	$N = 250$	exact Mittag–Leffler	8.70×10^{-4}
Caputo PECE	$N = 500$	exact Mittag–Leffler	4.64×10^{-4}
Caputo PECE	$N = 1000$	exact Mittag–Leffler	2.52×10^{-4}
Fractional-delay, $\alpha = 1$	$N = 200$	DDE RK4	6.75×10^{-6}
Fractional-delay, $\alpha = 1$	$N = 400$	DDE RK4	1.68×10^{-6}
Fractional-delay, $\alpha = 1$	$N = 800$	DDE RK4	4.20×10^{-7}
Fractional-delay, $(\alpha, \tau) = (0.85, 0.35)$	$N = 200$	$N = 800$	2.18×10^{-3}
Fractional-delay, $(\alpha, \tau) = (0.85, 0.35)$	$N = 400$	$N = 800$	8.05×10^{-4}

The gates are deliberately heterogeneous. Two implementations sharing one discretization can agree while both are wrong; the informative comparisons are against mathematically different references. Hence the choices above: delegation (not duplication) at $\tau = 0$, and an independent integer-order solver at $\alpha = 1$, where the memory convolution degenerates. The two checks constrain opposite boundaries of (2.4).

4.4. Experimental grid and reproducibility

The benchmark uses a small set of interpretable axes (Table 3), all under the actuator budget $\|u\|_\infty \leq 0.10$. The four-class stable benchmark sweeps $A \in \{0.20, 0.25\}$, $\alpha \in \{0.70, 0.85, 0.90, 0.95\}$, three SNR levels, three observation channels, and six waveforms, at 200 replicates per cell (1512 cells; Section 8). The fractional-delay campaign is intentionally smaller: $A = 0.25$, $\alpha \in \{0.75, 0.85, 0.95\}$, $\tau \in \{0.15, 0.30, 0.45\}$ for the design comparison, and the 7×6 grid $\alpha \in \{0.70, \dots, 1.00\}$, $\tau \in \{0, \dots, 0.50\}$ for the bridge maps.

Table 3. Main numerical-design axes. The fractional-delay campaign is intentionally targeted and is separate from the larger four-class benchmark.

Component	Values / protocol
Peak input budget	$\ u\ _\infty \leq 0.10$ for all six waveform families
Four-class stable benchmark	ODE, Caputo, retarded DDE, latent-3; $A \in \{0.20, 0.25\}$; 200 replicates per cell
Fractional orders in benchmark	$\alpha \in \{0.70, 0.85, 0.90, 0.95\}$ where applicable
Observation channels	prey only, predator only, both species
Noise regimes	high, medium, and low SNR
Fractional-delay design grid	$A = 0.25$, $\alpha \in \{0.75, 0.85, 0.95\}$, $\tau \in \{0.15, 0.30, 0.45\}$
Fractional-delay bridge grid	$\alpha = 0.70, 0.75, \dots, 1.00$ and $\tau = 0, 0.10, \dots, 0.50$
Simulation horizon	$T = 12$ for the principal trajectory/design comparisons

Every figure quantity traces to a JSON or NPZ file written by the simulation scripts; the fractional-delay script emits its validation sequence, design ranking, and bridge arrays together with Figures 7–10. Separating simulation output from rendering allows plots to be redrawn without rerunning the archived 1512-cell benchmark, and the archived statistics to be audited without redrawing.

5. Frequency-Domain and Finite-Horizon Evidence

The structural results of Section 3 are visible in two frequency-domain plots, computed before any benchmark. Figure 4 compares $|G(i\omega)|$ on the prey channel at $A = 0.25$ for the four mechanisms; Figure 5 shows the phase, which on the collocated channel converges to $-\alpha\pi/2 \approx -76.5^\circ$ for $\alpha = 0.85$ — the scale-free fractional signature of Theorem 3.1.

5.1. From exact signatures to finite-horizon ambiguity

The two panels must be read jointly with Corollary 3.3. The $O(\omega^{-\alpha})$ decay and the $-\alpha\pi/2$ phase identify the Caputo channel asymptotically, but an experiment observes a finite band $[\omega_{\min}, \omega_{\max}]$, a finite window $[0, T]$, and noise

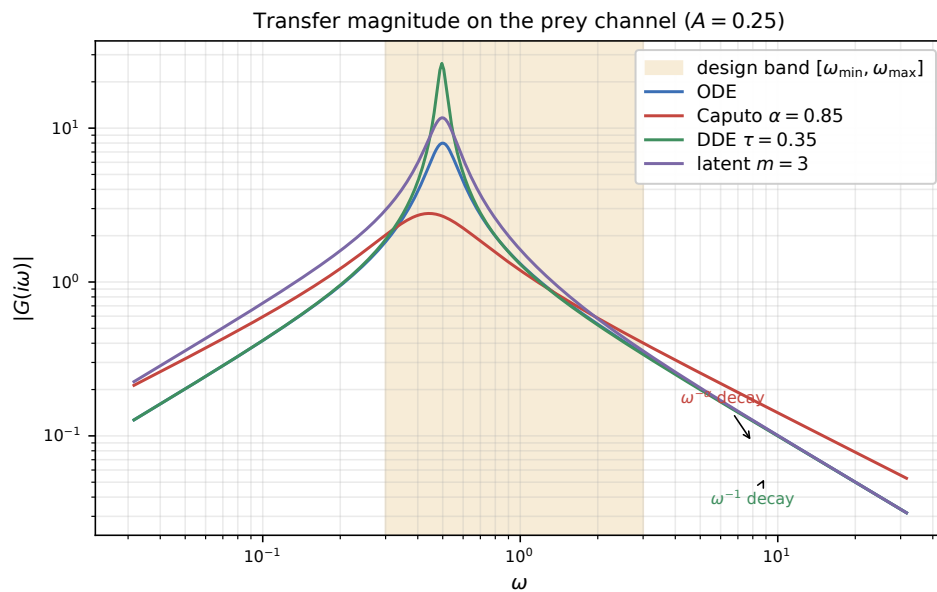


Figure 4. $|G(i\omega)|$ on the prey channel at $A = 0.25$ for ODE, Caputo ($\alpha = 0.85$), DDE ($\tau = 0.35$), and latent ($m = 3$). The Caputo curve separates over the design band and approaches its $O(\omega^{-\alpha})$ decay.

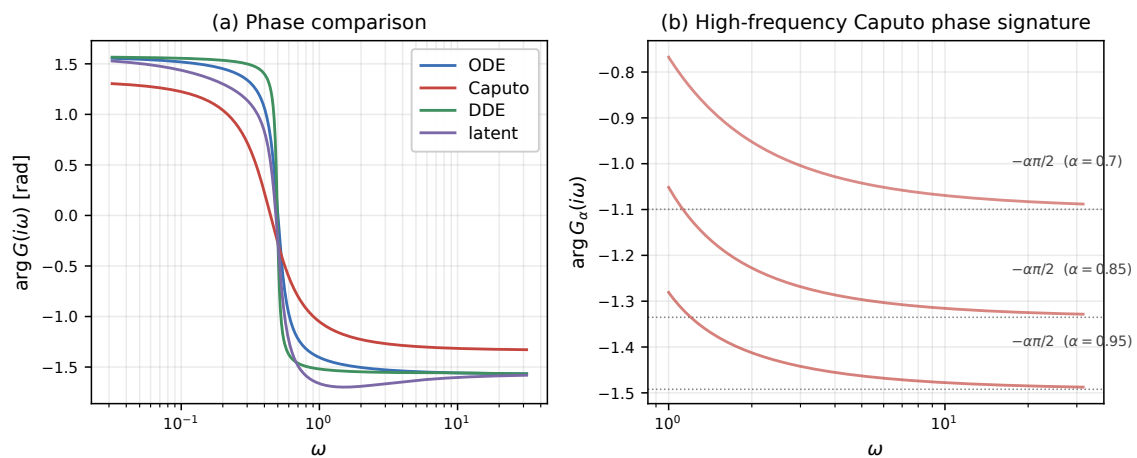


Figure 5. Phase on the prey channel. Left: the four mechanisms across frequency. Right: convergence of the Caputo phase to the asymptote $-\alpha\pi/2$, independent of scale.

of level σ . On that restricted set a latent model with enough modes matches the observable part of the response while remaining analytically distinct — the coexistence of exact separation (Theorem 3.1) and finite-horizon mimicry (Corollary 3.3) stated in frequency-domain terms.

The prey-response computation restates the point in the instrumented channel. Table 4 lists the certified L^1 bounds $\hat{E}_m^{\text{state}}$ and the induced testing floors at $\alpha = 0.85$, $T = 12$, $\sigma = 0.10$: from $m = 4$ to $m = 32$ the enclosure shrinks by a factor of 77 (from 0.5334 to 0.0069 at $A = 0.25$) and the testing bound rises from 0.374 to 0.498, i.e., to within 0.002 of the equal-prior chance level. Two bounds are reported. Since the two hypotheses are Gaussian with common covariance $\sigma^2 I$, the exact minimax error $\Phi(-d/2)$ with $d = \|\Delta\mu\|_2/\sigma$ is available and is uniformly tighter than the Pinsker bound; the difference matters only at small m , where Pinsker understates the floor by 0.034 at $m = 4$. We quote the exact bound and retain Pinsker for comparability with the literature.

Table 4. Certified finite-state approximation of the fractional prey response and induced testing lower bounds at $\alpha = 0.85$, $T = 12$, pulse $\|u\|_2 \leq 0.120$, and $\sigma = 0.10$. The exact bound is $\Phi(-d/2)$ with $d = \|\Delta\mu\|_2/\sigma$, valid because the two hypotheses share the covariance $\sigma^2 I$; Pinsker is reported for comparison and is uniformly looser.

m	$A = 0.25$			$A = 0.30$		
	$\hat{E}_m^{\text{state}}$	P_e^* (exact)	P_e^* (Pinsker)	$\hat{E}_m^{\text{state}}$	P_e^* (exact)	P_e^* (Pinsker)
4	0.5334	0.3745	0.340	0.4706	0.3888	0.359
8	0.1254	0.4700	0.462	0.0898	0.4785	0.473
16	0.0273	0.4935	0.492	0.0204	0.4951	0.494
32	0.0069	0.4983	0.498	0.0066	0.4984	0.498

Because Table 4 stops at $m = 32$ it shows only part of the obstruction. Table 5 continues it to $m = 128$ using an explicit positive-weight exponential-sum surrogate constructed as in Lemma 3.2 rather than the interval-optimised class $\mathcal{G}_m$. Because that surrogate is one admissible element and not the infimum, its error exceeds $\hat{E}_m^{\text{state}}$ — by factors 2.4 at $m = 4$ and 3.3 at $m = 32$ where the two overlap — so the induced floors are valid and conservative. They reach 0.4996 at $m = 64$ and 0.49999 at $m = 128$: at attainable latent orders the finite-state rival is indistinguishable from the fractional model under this protocol, and no experiment restricted to the declared budget can separate them.

Table 5. Prey-response L^1 error of an explicit positive-weight exponential-sum surrogate (not the certified infimum, hence an upper bound on $\hat{E}_m^{\text{state}}$) and the induced exact testing floor, $\alpha = 0.85$, $A = 0.25$, $T = 12$, $\|u\|_2 \leq 0.120$, $\sigma = 0.10$.

m	$\ g_\alpha - g_m\ _{L^1(0,T)}$	quadrature error	P_e^* (exact)
4	$1.29 \cdot 10^0$	$2.1 \cdot 10^{-7}$	0.2202
8	$4.98 \cdot 10^{-1}$	$1.0 \cdot 10^{-7}$	0.3827
16	$1.30 \cdot 10^{-1}$	$7.6 \cdot 10^{-9}$	0.4689
32	$2.31 \cdot 10^{-2}$	$3.0 \cdot 10^{-8}$	0.4945
64	$1.58 \cdot 10^{-3}$	$2.1 \cdot 10^{-8}$	0.4996
128	$4.83 \cdot 10^{-5}$	$6.3 \cdot 10^{-9}$	0.49999

The table is one-sided. A large lower bound on P_e^* proves that the specified experiment cannot reliably separate g_α from at least one admitted rival g_m ; a small bound proves nothing in the converse direction, so such cells are inconclusive rather than certified-easy. Figure 6 applies this reading across (α, A, m) : cells outside the focal Matignon-stable regime ($\alpha > \alpha^*(A)$) are hatched and excluded from the headline interpretation.

6. Fractional-Delay Model: Dedicated Simulation Study

Sections 3–5 treated fractional memory and explicit delay as competing hypotheses. This section studies the mechanism carrying both, via the combined model (2.4). Its boundaries coincide with the two mechanisms already characterized: $\tau = 0$ is the pure Caputo model and $\alpha = 1$ is the DDE benchmark, with both limits verified to the tolerances of Table 2. The interior of $(\alpha, \tau) \in [0.70, 1] \times [0, 0.5]$ then measures whether the two effects reinforce, cancel, or generate responses outside the reach of either pure explanation.

The scope is deliberately limited: no general theory of fractional delay equations is attempted. The study is a stress test of the discrimination framework under identical conditions to the rest of the paper — same ecology, same prey channel, same budget $\|u\|_\infty \leq 0.10$, same observables — with only (α, τ) varying. What it exposes is invisible to pairwise comparisons: when a system carries a hereditary response *and* a physiological lag, a binary “fractional versus delay” classification is misspecified even if each limiting model fits part of the data.

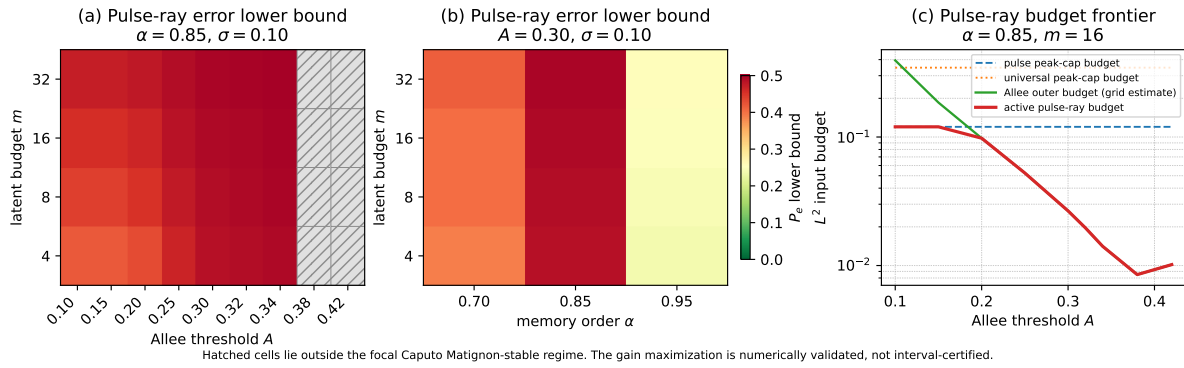


Figure 6. Protocol-restricted discrimination atlas over (α, A, m) : lower bound on the model-selection error under the declared pulse protocol. The bound increases with latent budget m and toward weakly separated regimes; hatched cells violate the Matignon condition $\alpha < \alpha^*(A)$ and are excluded from stable-regime conclusions.

6.1. Transient trajectories

Figure 7 fixes $A = 0.25$, $\alpha = 0.85$, applies the identical prey pulse, and varies $\tau \in \{0, 0.15, 0.35, 0.50\}$. The prey trajectory changes little; the predator response shifts visibly with τ and its late transient becomes increasingly delay-sensitive. All four trajectories satisfy $x(t) > A$ throughout, with minimum margin ≥ 0.356 (Table 6).

6.2. Fractional-order/delay parameter plane

Figure 8 maps the 7×6 grid $\alpha \in \{0.70, \dots, 1.00\}$, $\tau \in \{0, \dots, 0.50\}$. The edge $\tau = 0$ has zero distance to the pure Caputo model and the edge $\alpha = 1$ zero distance to the DDE, by construction; the quantity of interest is the interior escape rate from both. The right panel adds the safety dimension: a delay that leaves the margin nearly untouched at $\alpha = 0.75$ (from 0.3917 to 0.3850 over $\tau \in [0.1, 0.5]$) removes 21% of it at $\alpha = 0.95$ (from 0.3345 to 0.2629).

Table 6 lists nine representative cells. For each fixed α , d_{Caputo} increases monotonically in τ while d_{DDE} remains large (≥ 0.16 in every displayed cell). The steepest row is $\alpha = 0.95$: d_{Caputo} grows by a factor of 9.4 (from 0.0299 at $\tau = 0.10$ to 0.2805 at $\tau = 0.50$) while the minimum margin falls from 0.3345 to 0.2629. In this corner of parameter space, added delay increases identifiability and ecological risk simultaneously.

Table 6. Representative cells from the fractional-delay bridge at $A = 0.25$ under the pulse experiment. Distances are trajectory L^2 distances to the corresponding limiting models.

α	τ	distance to Caputo	distance to DDE	min. $[x(t) - A]$
0.75	0.10	0.0063	0.3426	0.3917
0.75	0.30	0.0182	0.6145	0.3887
0.75	0.50	0.0315	1.0086	0.3850
0.85	0.10	0.0103	0.2900	0.3724
0.85	0.30	0.0334	0.5683	0.3655
0.85	0.50	0.0625	0.9606	0.3566
0.95	0.10	0.0299	0.1645	0.3345
0.95	0.30	0.1142	0.4330	0.3123
0.95	0.50	0.2805	0.6875	0.2629

6.3. Which input exposes the combined mechanism?

Each input is scored over the nine cells $\alpha \in \{0.75, 0.85, 0.95\} \times \tau \in \{0.15, 0.30, 0.45\}$ by the mean of $\min(d_{\text{Caputo}}, d_{\text{DDE}})$; the min penalizes inputs that separate the combined response from only one limit. Results (Figure 9, Table 7): multisine attains the highest score (0.1332) with crossing fraction 1/3; PRBS (0.0909), sinusoid (0.0894), and chirp (0.0861) cross in all nine cells; pulse (0.0627) and multiscale (0.0338) never cross but discriminate weakly. The safety-informativeness tension of Section 8 reappears unchanged in the two-memory model.

6.4. Delay sweep

Figure 10 sweeps $\tau \in [0, 0.55]$ at twelve points for $\alpha \in \{0.75, 0.85, 0.95\}$. In all three orders the predator peak time increases with τ , its amplitude grows moderately (by $< 1\%$), and the minimum Allee margin decreases — weakly

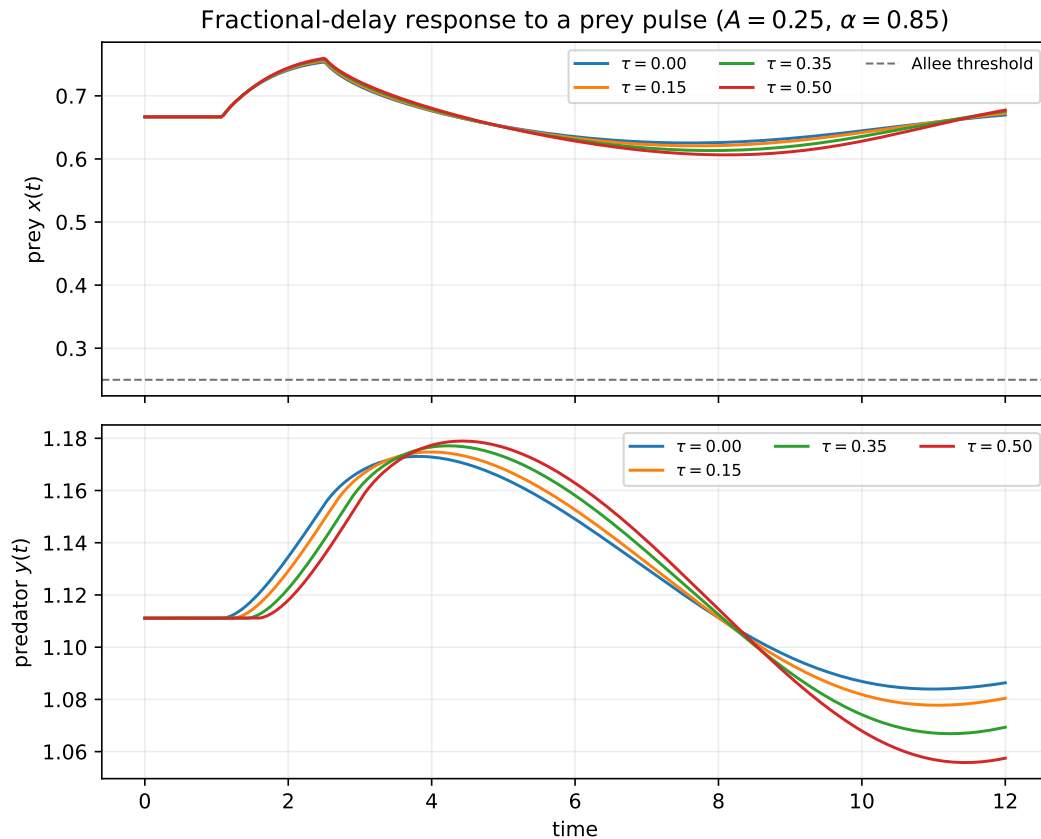


Figure 7. Combined fractional-delay response to the same prey pulse at $A = 0.25$ and $\alpha = 0.85$. Increasing delay shifts the predator response and reshapes the late transient while the fractional-memory relaxation pattern persists.

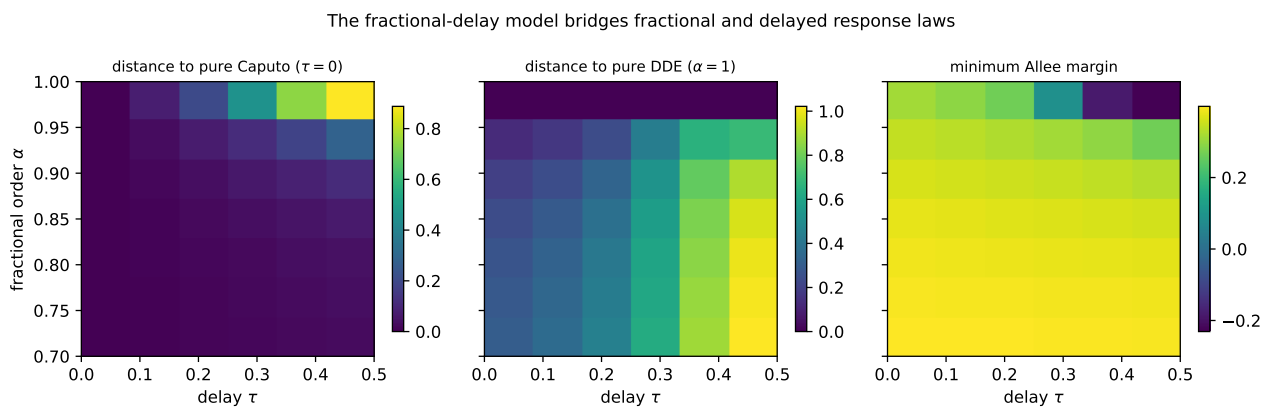


Figure 8. Fractional-delay bridge over (α, τ) . Left: trajectory distance to the pure Caputo limit. Center: distance to the pure DDE limit. Right: minimum Allee margin under the same pulse. Parts of the interior are poorly represented by either limit alone.

Table 7. Fractional-delay design stress test at $A = 0.25$. The score averages the minimum distance to the Caputo and DDE limits over $\alpha \in \{0.75, 0.85, 0.95\}$ and $\tau \in \{0.15, 0.30, 0.45\}$.

Input	mean bridge score	worst Allee margin	crossing fraction
multisine	0.1332	-0.2103	0.333
PRBS	0.0909	-0.2814	1.000
sinusoid	0.0894	-0.2777	1.000
chirp	0.0861	-0.2763	1.000
pulse	0.0627	0.2770	0.000
multiscale	0.0338	0.3171	0.000

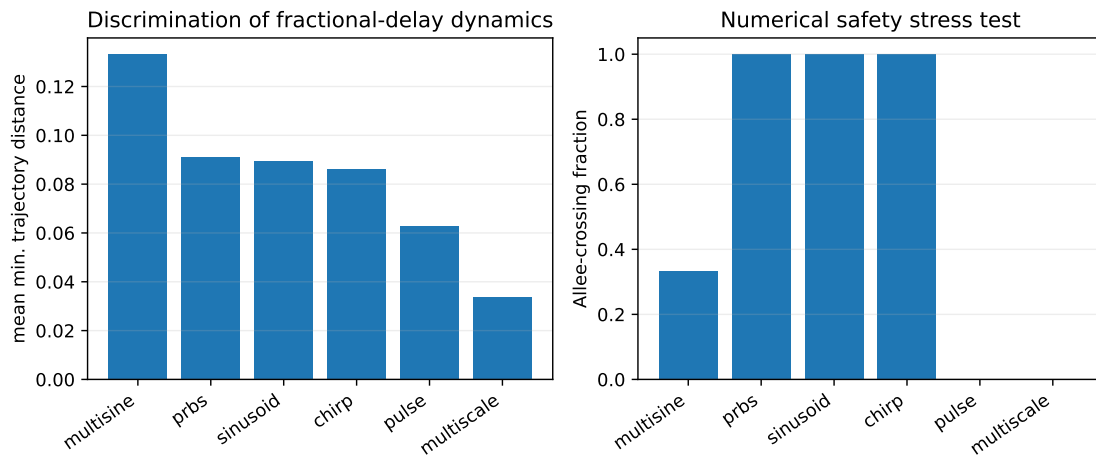


Figure 9. Input-design stress test for fractional-delay dynamics. Left: mean minimum trajectory distance from the two limiting explanations (pure Caputo and pure DDE). Right: numerical Allee-crossing fraction on the same (α, τ) grid. Broadband inputs expose the combined mechanism more strongly and are substantially less safe.

at $\alpha = 0.75$, strongly at $\alpha = 0.95$. Delay is thus observable as a reproducible timing shift, not merely as a fitting parameter, and the shift carries a quantifiable safety cost.

Endpoints (Table 8): at $\alpha = 0.85$, the change $\tau : 0 \rightarrow 0.55$ moves the predator peak from $t = 3.814$ to $t = 4.500$ (+0.686) and the margin from 0.3759 to 0.3548 (−5.6%); at $\alpha = 0.95$ the same change moves the peak from $t = 4.157$ to $t = 4.886$ and the margin from 0.3434 to 0.2384 (−30.6%).

Table 8. Endpoint summary of the delay sweep under the pulse experiment. Values compare $\tau = 0$ with $\tau = 0.55$.

α	τ	predator peak time	predator peak	min. Allee margin
0.75	0	3.343	1.1634	0.3934
0.75	0.55	4.029	1.1681	0.3843
0.85	0	3.814	1.1724	0.3759
0.85	0.55	4.500	1.1788	0.3548
0.95	0	4.157	1.1838	0.3434
0.95	0.55	4.886	1.1925	0.2384

The sampling-design implication is direct. Delay is expressed in the timing of the predator peak (at $t \approx 3.3$ –4.9 across the tested grid); fractional memory in the relaxation tail beyond it. A window restricted to the initial prey perturbation misses both signatures: it ends before the delayed predator response and far before the tail. Dense sampling around the predicted peak, extended through the tail, captures both — the recommendation quantified in Section 9.

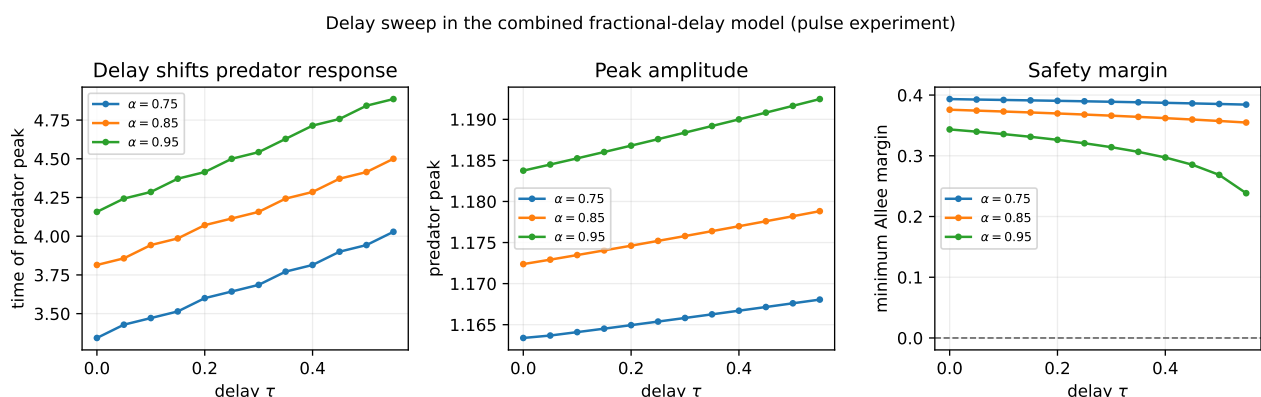


Figure 10. Delay sweep under the pulse experiment. The predator peak arrives later as τ grows, its amplitude increases moderately, and the numerical Allee margin shrinks — consistently across the three displayed fractional orders.

7. Experiment Design and Safety

The design theory itself is classical in outline; we state just enough here to motivate the waveforms, and refer the full development with proofs to the Supplementary Material. For two Gaussian linearized models with common covariance, the pairwise information carried by an input u takes the quadratic form

$$D_{ij}(u) = \frac{1}{2} \langle u, Q_{ij} u \rangle,$$

so under the energy constraint $\|u\|^2 \leq E$ the optimum is the leading eigenfunction of Q_{ij} , and the robust frequency-domain problem admits an optimum with finite frequency support (both proved in the Supplementary Material). These two facts justify restricting the search to the six implementable waveform families of Figure 11 instead of an infinite-dimensional input space.

Six candidate designs at a common peak-amplitude budget $\|u\|_\infty = 0.10$, grouped by safety behavior

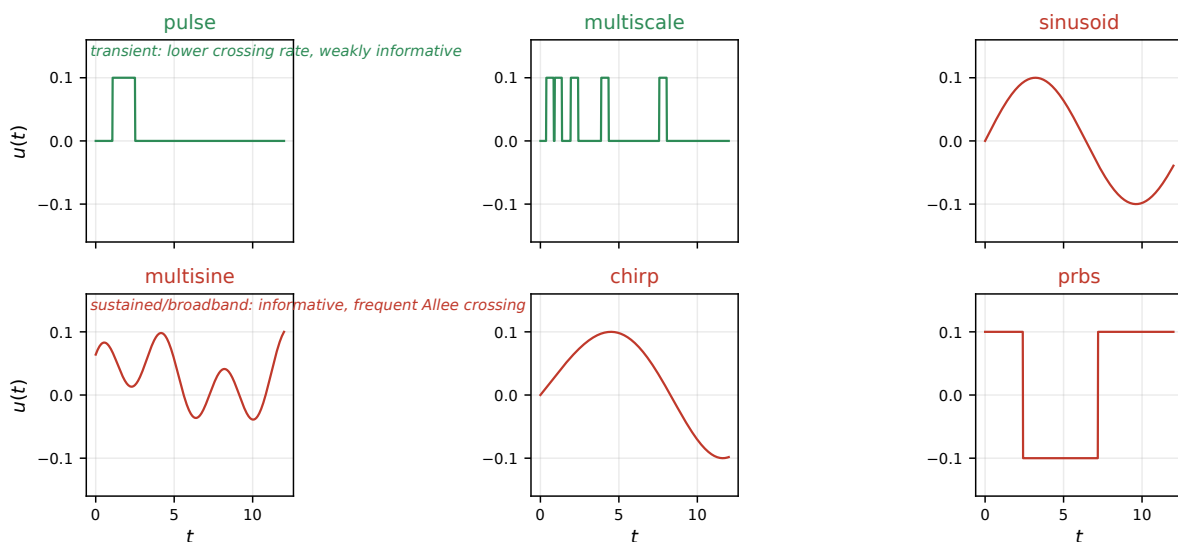


Figure 11. Six candidate perturbations at the common peak-amplitude budget. Transient and broadband designs interrogate the memory law in different ways — and behave very differently with respect to safety once the nonlinear simulations run.

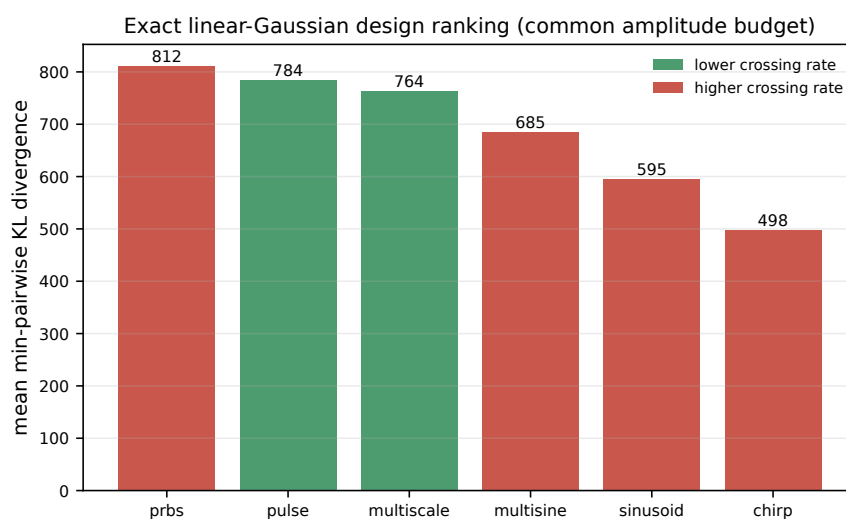


Figure 12. Linear-Gaussian design ranking by mean minimum pairwise KL divergence. Nonlinear state safety is invisible at this level, so the ranking serves as an information benchmark, not as a final experimental recommendation.

7.1. Strong-Allee safety

The one-sided Allee constraint requires $x(t) > A + \eta$ for all $t \in [0, T]$. For the linearized system, a sufficient pulse-amplitude envelope follows from the prey impulse gain; for the nonlinear system, the strict inward-pointing rectangle theorem and its four face inequalities are given in the Supplementary Material. One structural point belongs in the main text: state safety alone does *not* bound input energy. A stable strictly proper channel attenuates high frequencies as $O(\omega^{-1})$ or faster, so an oscillatory input of arbitrarily large $\|u\|_2$ can leave the state nearly unmoved. Every quantitative testing bound in this paper is therefore tied to an explicit actuator or protocol class — here, the declared peak-amplitude pulse budget $\|u\|_\infty \leq 0.10$, $\|u\|_2 \leq 0.120$.

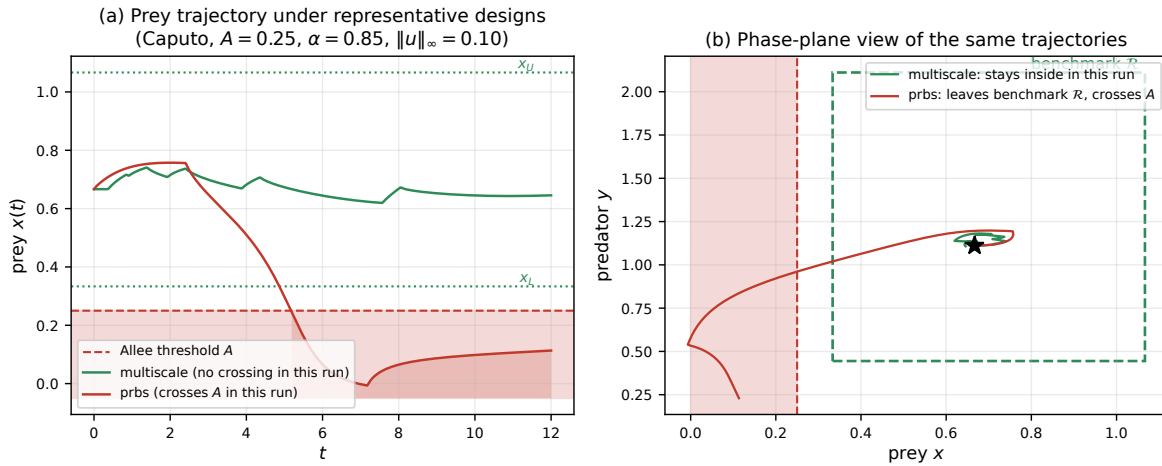


Figure 13. Representative nonlinear safety geometry. In the run shown, a multiscale transient stays above the Allee threshold while PRBS crosses it. The rectangle drawn is a benchmark diagnostic region; rigorous positive invariance requires the strict face conditions stated in the Supplementary Material.

8. Large-Scale Model-Discrimination Benchmark

The benchmark compares ODE, Caputo, retarded DDE, and latent-3 mechanisms at matched operating conditions. The primary stable-regime task uses $A \in \{0.20, 0.25\}$; a separate task at $A > 2/7$ addresses the easier stability-verdict question. The decision rule is BIC throughout — an information criterion, not Bayesian model evidence, and it is reported as such.

8.1. Benchmark design and decision rule

The benchmark is not organized around any single synthetic trajectory. It asks whether the qualitative design conclusions survive joint variation of memory strength (α), observation noise (three SNR levels), observation channel (three), and waveform (six families). For each generated trajectory, all four candidates are fitted under one observation protocol and compared by BIC; the object under evaluation is a reproducible decision rule.

Two factorial layers are kept separate. The nonlinear stable-regime benchmark has 1512 cells $\times$ 200 stochastic replicates (302,400 trajectories, zero solver divergences); a linear-Gaussian factorial of 1458 cells serves only as an information-ranking reference (Figure 12). The layers can disagree: the linear ranking measures local operator separation, while the nonlinear layer sees the Allee geometry, and an input that tops the linear ranking is disqualified once its trajectories cross $x = A$ (Table 10).

Macro-averaged accuracy in the primary four-class stable task is 0.537 versus chance 0.25. By input family:

$$\begin{aligned} \text{PRBS (0.584)} &\approx \text{multisine (0.583)} > \text{sinusoid (0.534)}, \\ &> \text{chirp (0.507)} > \text{pulse (0.414)} > \text{multiscale (0.324)}. \end{aligned}$$

Difficulty is dominated by two axes (Table 9). As $\alpha \rightarrow 1$ the mechanisms converge and accuracy falls from 0.702 ($\alpha = 0.70$) to 0.312 ($\alpha = 0.95$); across noise, accuracy is 0.756/0.444/0.273 at high/medium/low SNR. The channel effect is smaller: both-species observation attains 0.530, prey-only 0.516 (the directly actuated channel remains structurally sufficient), predator-only 0.427.

Table 9. Stable-regime four-class BIC accuracy stratified by fractional order, SNR, and observation channel.

Axis	level	accuracy	Axis	level	accuracy
α	0.70	0.702	SNR	high	0.756
	0.85	0.557		medium	0.444
	0.90	0.467		low	0.273
	0.95	0.312	channel	prey	0.516
				predator	0.427
				both	0.530

Class-wise statistics refine the picture (Figure 14). ODE: recall 0.972, precision 0.227 — under weak excitation the subtler mechanisms are absorbed into the simplest class. Caputo: recall 0.524, precision 0.984 — a Caputo verdict, once issued, is nearly always correct. DDE: 0.410/0.731; latent-3: 0.242/0.848. The dominant failure mode is therefore conservative collapse toward the simpler explanation when the data underdetermine the memory law, not symmetric confusion among mechanisms.

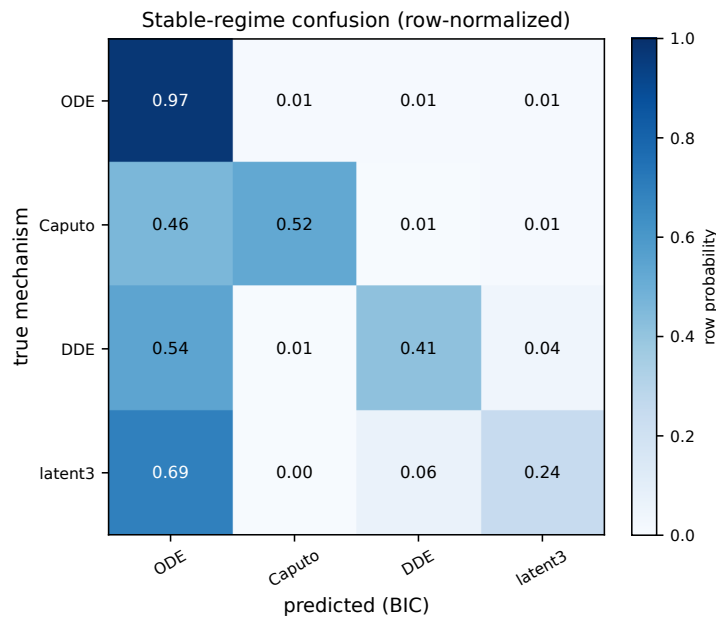


Figure 14. Row-normalized confusion matrix for the stable four-class task. Under weak excitation the simplest ODE mechanism absorbs the rest; Caputo predictions, when made, are highly precise.

8.2. The trade-off within classical waveform families

Numerical state safety is reported on its own axis, never folded into the information score. At the common budget $\|u\|_\infty \leq 0.10$: multiscale never crosses the threshold (rate 0.000) and discriminates worst (0.324); PRBS and sinusoid discriminate best (0.584, 0.534) and cross at rates 1.000; multisine matches PRBS in accuracy (0.583) at crossing rate 0.357.

Table 10 gives the frontier *within the six classical families of Section 7*. Section 8.4 shows that this frontier is a property of that parameterisation and not of the system. Multiscale retains a positive minimum margin of 0.292 at the lowest accuracy; the pulse is the practical compromise, crossing at rate 0.071 with accuracy 0.414 — below the broadband designs by 0.17.

The table also fixes a methodological point: the linear-Gaussian ranking alone cannot select the final experiment. Linear theory scores an input by operator separation; the nonlinear ecology can answer the same input by leaving the coexistence basin, after which no local identification experiment remains. Safety must be evaluated on the nonlinear state trajectory, not extrapolated from the linear information score.

8.3. Amplitude is the wrong knob

Before accepting the frontier of Table 10 one must ask whether it is simply a symptom of excessive excitation. It is not. Repeating the entire benchmark at peak amplitudes 0.050 and 0.063 alongside the reference 0.100 (2268 cells

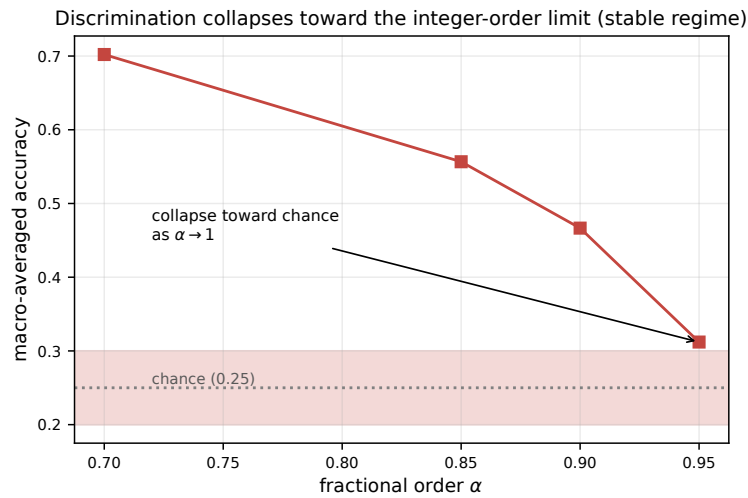


Figure 15. Model-selection accuracy versus fractional order. The closer α gets to the integer-order limit, the harder the memory mechanisms are to tell apart.

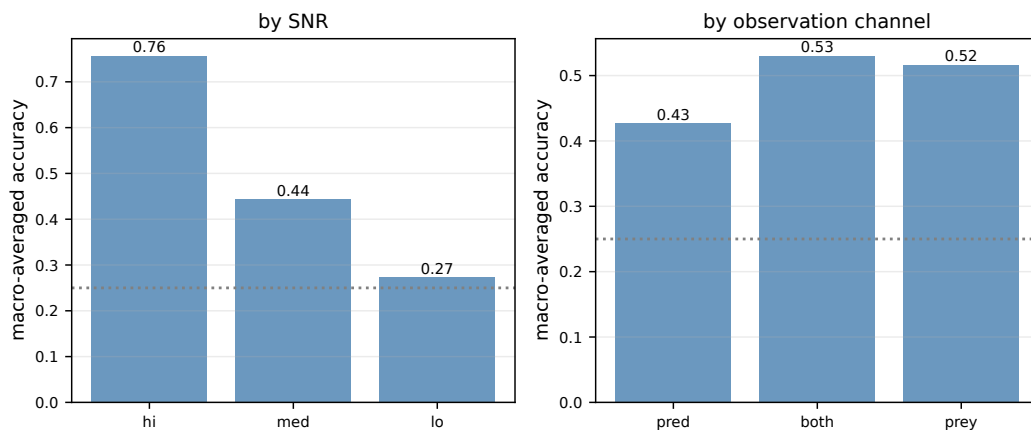


Figure 16. Accuracy stratified by SNR and observation channel. The benchmark is strongly noise-limited; observing both species gives the best average accuracy.

Table 10. Nonlinear stable-benchmark trade-off at the common peak-amplitude budget. Safety values are numerical benchmark diagnostics, not invariant-set certificates.

Input	macro accuracy	Allee crossing rate	minimum Allee margin
PRBS	0.584	1.000	-0.3139
multisine	0.583	0.357	-0.2405
sinusoid	0.534	1.000	-0.3058
chirp	0.507	0.929	-0.3012
pulse	0.414	0.071	-0.1207
multiscale	0.324	0.000	0.2921



Figure 17. Trade-off across the six classical waveform families. Broadband and sustained designs climb in discrimination but drift toward higher Allee-crossing rates; transient designs stay in the low-crossing, less-informative corner. The frontier shown is specific to these families: the piecewise-constant designs of Section 8.4 lie above and to the left of every point plotted here.

in total, zero divergences) leaves the ordering of designs unchanged and does not restore safety: PRBS remains the most informative design at every amplitude while still crossing the threshold in 71.4% of cells at half amplitude (92.9% at 0.063, 100% at 0.100). Macro-accuracy falls monotonically with amplitude (0.457, 0.471, 0.537), so the low-amplitude runs are reported as linear-certificate diagnostics, not as a safe benchmark.

The realised response gain $\max_t |z(t) - z^*|/\|u\|_\infty$ makes the mechanism explicit. For the crossing designs this gain *increases* as the amplitude decreases, reaching 28.07 at amplitude 0.050 against the linear certificate $\Gamma_T = 5.78$, a violation by a factor of 4.86; for the two designs that never cross it stays between 1.47 and 2.06 and is essentially amplitude-independent, as a linear response must be. Once a trajectory leaves the Allee basin, $\max_t |z - z^*|$ is set by the basin geometry rather than by the input, so the ratio scales like $1/\|u\|_\infty$. The linear certificate therefore cannot be used to justify safety for any crossing design at any of the three amplitudes, and shrinking the input does not convert an unsafe design into a safe one.

8.4. Waveform search: the frontier is a parameterisation artifact

If amplitude is not the knob, the waveform shape may be. We therefore searched for inputs that are safe *and* informative outside the six classical families, at the same peak-amplitude budget $\|u\|_\infty \leq 0.100$ and under the hard constraint that the minimum Allee margin over all four candidate models be at least $\delta = 0.05$. The search covers piecewise-constant inputs with 4, 6 and 8 levels, free multisines with 3–6 harmonics, and pulse trains with 2–4 pulses, scored by the minimax pairwise separation $J(u) = \min_{a \neq b} \|\mu_a(u) - \mu_b(u)\|/\sigma$ over the four classes; 163 840 candidates were evaluated by Sobol sampling with Nelder–Mead refinement of the per-family best, and 87 601 of them (53.5%) satisfy the safety constraint. Re-scored under the same objective, exactly one of the six classical designs is safe at this amplitude (multiscale, $J = 1.45$) and it is the least informative of the six; the best safe candidate found reaches $J = 6.23$, a factor of 4.3 higher, and every one of the twelve best safe candidates is piecewise constant — a class the six families do not contain.

Validating the six leading candidates with the same BIC classifier, the same 56-cell grid and the same replicate seeds as Table 10 confirms the surrogate (Table 11). All six are safe, with zero observed crossings and verified margins between +0.067 and +0.176, and five of the six outrank the best classical design of *any* kind, safe or not. The leading design attains macro-accuracy 0.803 against 0.514 for the only safe classical design and 0.763 for multisine, which crosses in 35.7% of cells. The gain is concentrated exactly where the safe classical design fails: Caputo recall rises from 0.404 to 0.911 and DDE recall from 0.445 to 0.853.

Safety status of these trajectories is stated exactly. Each is verified a posteriori: the solver output is refined until the

Table 11. Waveform search validated with the benchmark classifier at $\|u\|_\infty \leq 0.100$: 56 cells $\times$ 100 replicates per design, identical seeds, no divergences. *pwck* denotes a k -level piecewise-constant input on equal sub-intervals.

design	family	macro	micro	crossing rate	min. margin	Caputo	DDE
S0	pw6	0.803	0.787	0.000	+0.092	0.911	0.853
S3	pw6	0.783	0.753	0.000	+0.122	0.910	0.927
S2	pw6	0.783	0.765	0.000	+0.096	0.875	0.816
S1	pw6	0.780	0.744	0.000	+0.109	0.838	0.897
S5	pw8	0.769	0.727	0.000	+0.176	0.844	0.926
multisine	classical	0.763	0.725	0.357	−0.240	0.844	0.902
S4	pw6	0.750	0.720	0.000	+0.067	0.766	0.743
prbs	classical	0.711	0.713	1.000	−0.314	0.802	0.521
sinusoid	classical	0.685	0.687	1.000	−0.306	0.844	0.549
chirp	classical	0.683	0.684	0.929	−0.301	0.832	0.547
pulse	classical	0.621	0.552	0.071	−0.121	0.528	0.647
multiscale	classical	0.514	0.444	0.000	+0.292	0.404	0.445

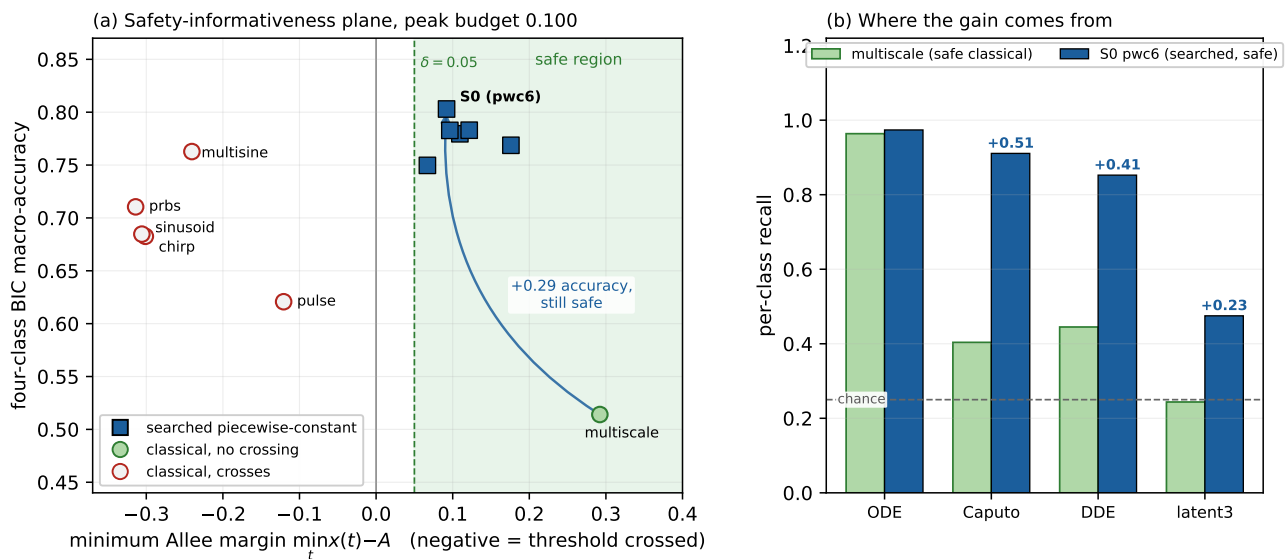


Figure 18. The safety-informativeness plane at the common peak-amplitude budget $\|u\|_\infty \leq 0.100$. (a) Each point is one design: minimum Allee margin over the four candidate mechanisms against four-class BIC macro-accuracy (56 cells $\times$ 100 replicates, identical seeds). Circles are the six classical waveform families, filled where no crossing occurs; squares are the leading designs returned by the constrained search over piecewise-constant, free-multisine and pulse-train inputs. Of the twelve designs, only multiscale and the searched pw6 design are safe under *all four* mechanisms. The searched designs occupy a region of the plane that the classical families do not reach. (b) Per-class recall of the safe classical design against the leading searched design; the gain is concentrated on the Caputo and delayed mechanisms, which the safe classical design barely distinguishes from chance. The search establishes *existence* of such designs — Sobol sampling with local refinement over the stated families, at a single amplitude and a single α — and is not a certified global optimum.

minimum prey value stabilises, and the gap between mesh nodes is closed by a rigorous modulus-of-continuity bound derived from the Caputo mild solution, so no crossing can hide between samples. For the two headline designs under the integer-order and finite-latent mechanisms — four trajectories in total — a stronger statement holds: a validated enclosure built from a piecewise-cubic reference, an interval-bounded defect and a one-sided Gronwall estimate gives self-consistent error bounds of $4.6 \cdot 10^{-3}$ and $4.2 \cdot 10^{-2}$, hence rigorous lower bounds of 0.441 and 0.309 on the prey minimum against the threshold $A = 0.25$. The delayed and Caputo mechanisms are verified but not enclosed: the delayed defect stalls at first order because input discontinuities re-enter through the retarded argument, and the fractional defect is limited by the solver's own order. The word *rigorous* is used below only for the four enclosed trajectories.

A design found by search on one operating point could simply be overfitted to it. The validation grid answers this without extra computation, because the search was run at $A = 0.25$ only while the grid also contains $A = 0.20$, which is therefore fully held out. Splitting the 672 validation cells accordingly, the searched designs lose on average 0.060 macro-accuracy moving to the held-out threshold, while the six classical families — which were never optimised at all — lose on average 0.087. The best searched design still leads the best classical design on held-out cells (0.907 against 0.837), and its advantage over the safe classical design is larger out of sample (+0.486) than in sample (+0.255), because multiscale itself degrades sharply at the lower threshold. There is thus no evidence that the search advantage is an artifact of selecting and evaluating at the same operating point.

Two qualifications are required. The search establishes *existence* of safe designs above the classical frontier, not global optimality: it is a Sobol-plus-local-refinement search, and no certified global bound over the piecewise-constant class is claimed. And the constraint is active — the refined optimum sits at the declared margin $\delta = 0.05$, while multiscale sits at 0.292 — so a stricter margin buys a lower objective and the trade-off is relocated rather than abolished. What does not survive is the reading of Table 10 as a property of the ecology: the accessible frontier lies well above it. The residual obstruction is the latent one, on the independent grounds of Section 3.5; the leading design's 0.475 recall against the finite-latent rival is consistent with that obstruction, but since BIC penalises the higher-dimensional model it cannot be read as evidence for it.

9. Prospective Experimental Interpretation

Laboratory predator–prey microcosms provide the controlled perturbation setting the simulations presume [2, 7, 26], and the numerical results translate into specific protocol constraints. A first experiment should operate in a stable strong-Allee regime ($A \leq 2/7$), apply a bounded prey perturbation ($\|u\|_\infty \leq 0.10$ in normalized units), observe both populations where instrumentation allows (accuracy 0.530 versus 0.516 prey-only; Table 9), and postpone broadband excitation until a transient probe has established the available margin.

The fractional-delay study adds one operational diagnostic. Delay moves the predator peak time (by up to +0.73 time units over $\tau \in [0, 0.55]$; Table 8); fractional order reshapes the long transient and phase. A scheme sampling densely around the predicted peak ($t \approx 3.3$ – 4.9 on the tested grid) and recording through the relaxation tail resolves both signatures; uniform sparse sampling can miss both.

A Bayesian sequential layer is specified in the Supplementary Material and deliberately not executed; the evidence in the main text is limited to deterministic solvers, interval enclosures, and BIC benchmarks, all rerunnable from the released scripts.

A two-stage prospective protocol

The observed frontier (Tables 10 and 7) argues against applying the most informative perturbation first. Stage 1 uses a transient input with low observed crossing rate (pulse, 0.071; multiscale, 0.000) to estimate the local response scale and verify a comfortable margin above A . Stage 2 is conditional: multisine-type excitation (accuracy 0.583, crossing rate 0.357) is applied only if Stage 1 leaves adequate margin. No Bayes-optimality is claimed for this sequencing; it is a direct reading of the frontier, summarized in Table 12.

Three measurement priorities follow from the simulations. (i) The actuated prey channel carries the strongest early signal: sample densely immediately after the perturbation. (ii) The delayed pathway is expressed in predator-peak timing: concentrate observations near the anticipated peak rather than uniformly. (iii) Fractional memory dominates the slow relaxation tail: extend the horizon well past the peak. The resulting design is nonuniform — dense around the forcing and the peak, sparser but longer through relaxation.

On units: the simulations are dimensionless with a normalized budget. A physical implementation must map state and input to organism counts, biomass, or resource-manipulation units and must fix an operational margin above the

Table 12. Prospective microcosm workflow suggested by the simulation results.

Stage	Perturbation	Primary measurements	Decision / purpose
0	no perturbation	baseline prey/predator variability	estimate noise and confirm operating point
1	pulse or multiscale	dense early prey observations; predator peak timing; minimum prey level	verify local response and available safety margin
2	multisine if margin permits	both species over the full transient and relaxation tail	maximize separation of memory mechanisms
3	optional repeat / altered horizon	same channels, replicated under matched conditions	test robustness of the selected mechanism and estimate variability

empirical Allee threshold. This calibration is application-specific and is not inferred from simulation here.

10. Discussion

Three conclusions are supported consistently by the analytical results of Section 3 and the numerical layers of Sections 5–8.

First, *structural identifiability and practical discrimination are different questions*. The Caputo transfer carries an exact $O(\omega^{-\alpha})$ signature (Theorem 3.1); finite latent models approximate the same response to any tolerance on $[0, T]$ (Corollary 3.3); and the testing bounds quantify the gap — $P_e^* \geq 0.498$ at $m = 32$ under the pulse protocol (Table 4). A flexible latent rival therefore remains problematic long after the model classes are proved analytically distinct.

Second, *delay and fractional memory should not be treated as mutually exclusive*. The simulations of Section 6 trace a continuous bridge between the Caputo limit ($\tau = 0$) and the DDE limit ($\alpha = 1$): delay governs predator-response timing, fractional order the relaxation structure, and interior cells sit far from both limits simultaneously ($d_{\text{Caputo}} = 0.28$, $d_{\text{DDE}} = 0.69$ at $(\alpha, \tau) = (0.95, 0.50)$; Table 6). Fitting the two endpoint models separately cannot reveal this; the composite stress test can.

Third, *within classical waveform families the most informative perturbation is not the safest*, by a wide margin: the accuracy-leading inputs cross the Allee threshold at rates 0.357–1.000 in the four-class benchmark (Table 10) and in 3/9 to 9/9 cells of the fractional-delay grid (Table 7), two independent measurements of the same effect. Crossing A is not a numerical artifact: it changes the basin structure, and the procedure ceases to be a local identification experiment at that point.

Fourth, and this qualifies the third conclusion decisively, *that trade-off belongs to the parameterisation and not to the ecology*. Neither reducing amplitude (Section 8.3) nor accepting the classical frontier is necessary: searching piecewise-constant inputs at the same amplitude budget under a hard margin constraint produces designs that are simultaneously safe and more informative than every classical design, safe or not — macro-accuracy 0.803 with zero crossings against 0.514 for the only safe classical family and 0.763 for the best classical design overall (Table 11). The frontier of Table 10 is therefore a statement about six waveform families, and reporting it as a property of safe ecological identification would be an error. What remains binding is the latent-complexity obstruction of Corollary 3.3, Theorem 3.4 and Tables 4–5: the induced testing floor reaches 0.49999 by $m = 128$, independently of the decision rule. The benchmark’s low recall against the finite-latent rival (0.475 for the leading safe design, and at or below chance for the class overall) is consistent with this, but BIC’s complexity penalty is a competing explanation that the present design does not separate, so the obstruction rests on Section 3.5 and not on the recall. The operative constraint on identifying ecological memory is the declared rival-complexity budget, not the safety margin.

Limitations. Theorem 3.5 is linearized and specific to the prey channel; a full-state, hierarchy-wide safety closure remains open. The waveform search of Section 8.4 establishes existence, not global optimality: no certified global bound over the piecewise-constant class is claimed, the search was run at a single amplitude and a single α , and its safety constraint is active at the declared margin. Safety of the leading designs is verified a posteriori by mesh refinement together with a rigorous inter-node bound from the Caputo modulus of continuity; this decides every trajectory tested but is not an interval enclosure of the solver error, and a validated enclosure was obtained for only four of the eight headline trajectories. The fractional-delay study is numerical only, with constant equilibrium history, one delayed trophic pathway, and the targeted grids of Table 3. The benchmark decides by BIC, not Bayesian evidence; adaptive design is specified (Supplementary Material) but not executed. This ledger separates proved results from computational evidence.

Relation to the previously submitted companion paper. The shared material is the strong-Allee backbone with its locked parameterization, from Ref. [14]; nothing else is shared. Ref. [14] concerns certification: computer-assisted verification of Caputo equilibria, the Matignon boundary $\alpha^*(A)$, extinction/boundedness, and the diagnostic failure of a forward-Euler surrogate. The present work concerns discrimination. Theorems 3.1, 3.4 and 3.5, Lemma 3.2, the six-family design study, the 1512-cell benchmark, and the fractional-delay simulations appear nowhere in Ref. [14]. Each inherited stability fact enters only to select an admissible operating regime, and is attributed where it enters.

What is genuinely added by the present paper. Backbone reuse is an experimental-design decision. With the ecology frozen, differences between simulated observations are attributable to the memory law or to the experiment, never to an altered population model. The paper starts where [14] ends: a certified operating point leaves open whether an observed transient identifies a fractional mechanism uniquely, whether a delayed or latent model can reproduce it (Corollary 3.3 says yes, at sufficient m), and which perturbations separate the alternatives within the safe regime. These questions require the mathematics of Section 3 and the computations of Sections 4–8, none of which appears in [14].

The figure sets reflect the same division: the centerpiece of [14] is a certified (A, α) stability atlas, whereas Figures 4–17 here concern response separation, approximation error, waveform design, BIC confusion, fractional-delay dynamics, and the safety–informativeness frontier. Only the model schematic (Figure 2) orients the reader in the common ecology; no certificate atlas or certificate count from [14] is used as evidence.

Relation to broader prior work. Safe information limits and safe active model discrimination exist as general theories [18, 22]; no novelty is claimed for safe experimental design per se. The contribution is the specific quantitative interaction: finite-horizon fractional approximation (Lemma 3.2, a borrowed construction), bounded latent complexity, strong-Allee geometry, and the combined fractional-delay stress test in a single frame.

Implications for experimental identification. Model selection here is a *controlled intervention problem*, not passive curve fitting. Under weak excitation, BIC collapses to the ODE explanation (recall 0.972, precision 0.227); stronger excitation separates the mechanisms and simultaneously raises the crossing rate. The fractional-delay study adds a distinct warning: both endpoint models can be individually identifiable while an interior composite occupies a response region neither represents (Table 6), so designs should test composite alternatives, not only the endpoints.

The operative objective is not “maximize information” but maximize information *subject to* a trajectory-level safety margin and a declared rival-complexity budget — and Section 8.4 shows that this constrained problem should actually be solved rather than approximated by choosing among textbook waveforms, since the loss from not solving it is 0.29 in macro-accuracy. The second constraint is forced by Corollary 3.3: an unrestricted latent hierarchy reproduces fractional memory on any finite horizon, so the allowed latent dimension m , delay range, forcing bandwidth, and observation horizon T belong in the scientific specification of the experiment, alongside organism and protocol.

11. Conclusion

The question posed at the outset — what would it take, experimentally, to establish that an ecological system carries fractional memory — receives a quantitative answer with one part negative and one part constructive.

The constructive part is that the apparent conflict between safety and information is largely an artifact of how the perturbation is parameterised. Across the six classical excitation families the picture is discouraging: the accuracy-leading designs leave the coexistence basin, and the single family that never does is the least informative of the six. Reducing amplitude does not resolve this, and the realised response gain shows why — once a trajectory escapes, its excursion is set by the basin geometry rather than by the input, so the linear certificate cannot license safety at any of the amplitudes tested. Optimising the waveform *shape* instead, over a piecewise-constant class the classical families do not contain and at the same peak-amplitude budget, recovers most of what the safe classical design loses: four-class accuracy 0.803 with no observed threshold crossings, against 0.514. Safety of these trajectories is verified a posteriori, and rigorously enclosed for four of the eight headline cases. A Pareto cost survives — the best unsafe candidate still separates the hypotheses better — so the correct statement is not that safety is free, but that the severe frontier previously read off the classical families is not a property of the ecology.

The negative part is independent of input design and does not yield to it. Computed directly at the prey-response level rather than inferred from the memory kernel, finite-state surrogates reach an L^1 error of $5 \cdot 10^{-5}$ at latent order 128, and the exact two-point minimax floor they induce reaches 0.49999 under the declared protocol — uniformly over

every admissible input, since the surrogate error is an operator norm. Structural distinctness (Theorem 3.1) therefore coexists with practical indistinguishability at attainable latent orders (Corollary 3.3). The benchmark is consistent with this, though there the model-selection penalty and the approximation-theoretic obstruction act in the same direction and are not separated, so the obstruction is asserted on the analytical grounds alone.

The practical consequence for experimental design is a change of question. Not *which model fits best*, nor even *which safe perturbation separates the mechanisms* — that one now has a constructive answer — but *what latent complexity the experiment is willing to admit as a rival*. The allowed latent dimension, delay range, forcing bandwidth and observation horizon belong in the scientific specification of the experiment alongside organism and protocol, because it is that declaration, and not the safety margin, that bounds what can be identified.

CRediT authorship contribution statement

Ibrahim Alraddadi: Conceptualization, Methodology, Formal analysis, Software, Investigation, Validation, Data curation, Visualization, Writing – original draft, Writing – review & editing.

Use of AI tools declaration

AI-assisted tools were used during manuscript preparation for language editing, LaTeX formatting, code and documentation assistance, and structural organization. Responsibility for the scientific content and the final submitted version remains with the author.

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The source code implementing the solvers, the waveform search, the four-class discrimination benchmark and the trajectory verification, together with the numerical artifacts from which every table and figure in this paper is generated, are available from the author on request.

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